Bioinformatics Update

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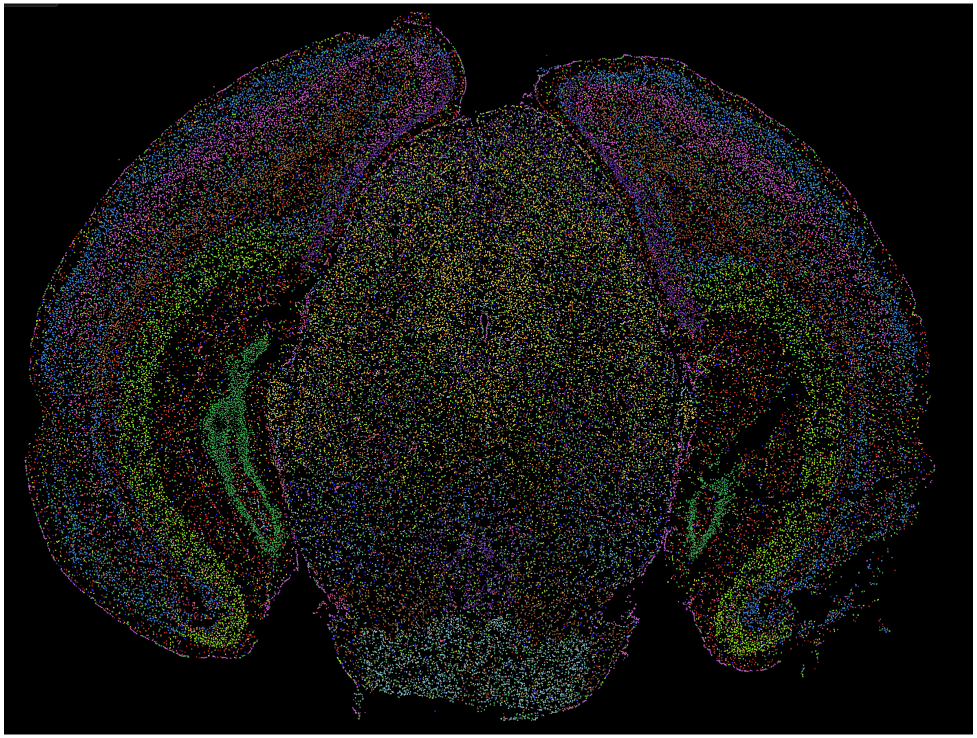
5/8/2025

Macaca mulatta (rhemac10)
MERFISH analysis

MERFISH Overview

- MERFISH = Multiplexed Error-Robust Fluorescence In Situ Hybridization
- Combines single-cell gene expression profiling with spatial information
- Enables mapping of RNA molecules directly in intact tissue slices
- Useful for mapping gene expression across tissues and discovering complex arrangement of cell types and states

[MERFISH Mouse Brain Receptor Map (Vizgen)]



Our MERFISH Project

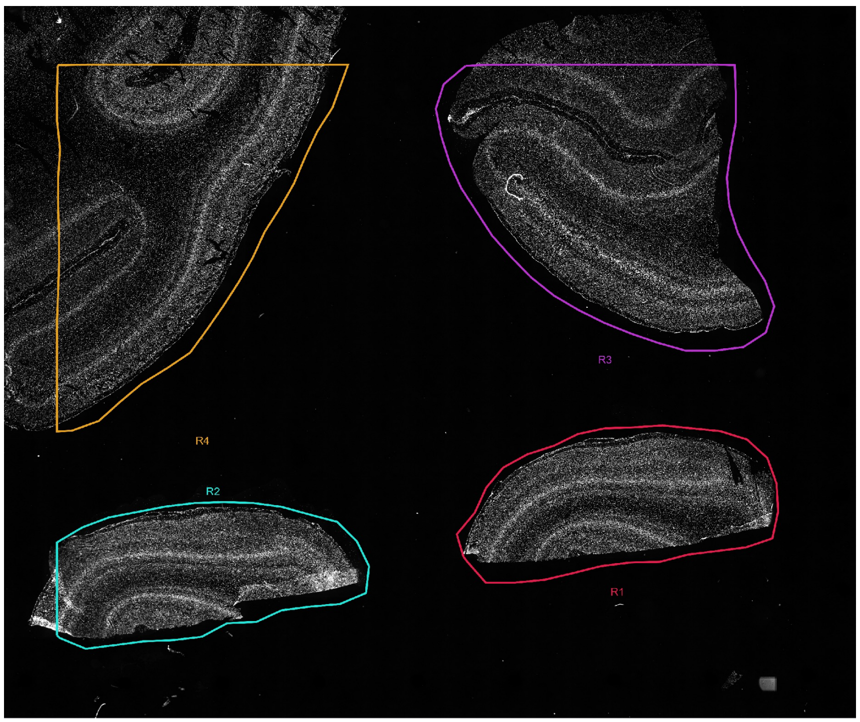
- Created a large data set (~6TB) in collaboration with the UMMS SCOPE facility & Vizgen on primary visual cortex (V1) of Rhesus macaque (*Macaca mulatta*) (rhemac10)
- Created a custom gene panel of 881 genes from rhesus macaque & human
    - Helpful for identifying different cell types: excitatory, inhibitory, glial, vascular, etc.), activity dependent genes, genes expressed in different diseases (ASD, SCZ)

Goals

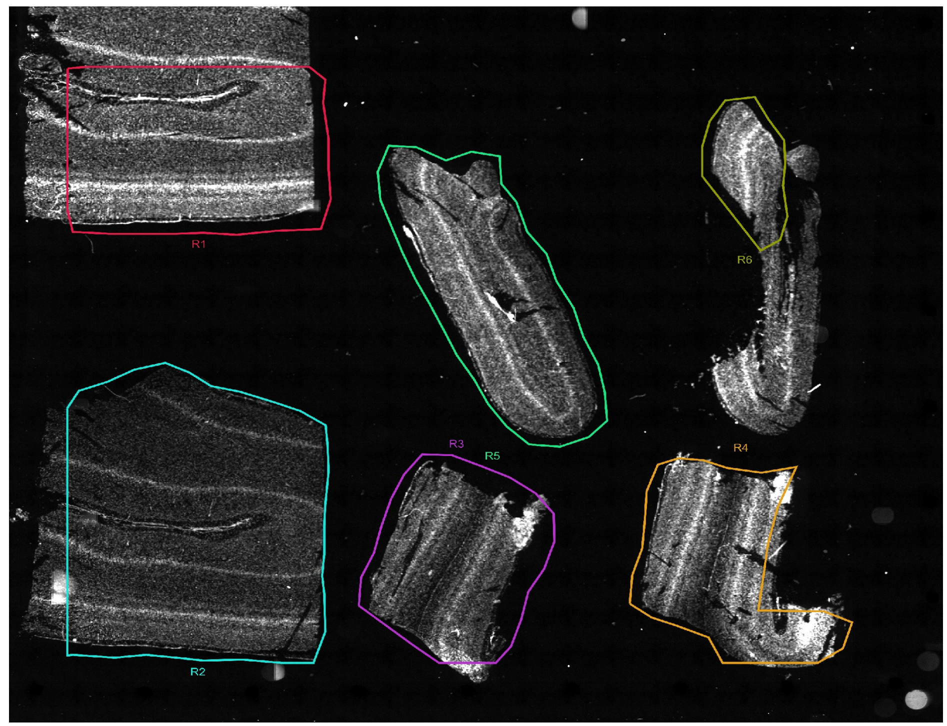
- Identify cell types in macaque V1
- Identify spatial organization of cell types
- Predict activity states in cells (active vs inactive)
- Compare gene expression data with and our rhemac10 snRNAseq dataset and Wei et al. (2022) to validate gene expression patterns in macaque V1

Samples

Slide 4 (R1, R2, R3, R4)



Slide 5 (R1, R2, R3, R4, R5, R6)



Vizgen HTML Reports

vizgen

MERSCOPE Analysis Report (202412041735_Boulting-Slide4-NHPBrain-DM1919_VMSC21910)

Sunday, December 08 at 09:08AM

Total transcripts detected: **194,976,417** Transcripts per 100 µm²: **105** Cell count: **317,665** Transcripts per cell – median: **286**

Overview Transcripts Cells Auxiliary stains

General

Statistics

Experiment area (mm²)	186.03
Total cell containing area (mm²)	183.09
Total cells detected	317,665
Total transcripts detected	194,976,417
Median transcripts per cell	286
Transcripts per 100µm² of transcript-containing tissue	105
Estimated tissue thickness (µm)	10.5
Tissue profile	flat

Gene Panel Metadata

Gene panel serial number	3
Gene panel name	Mulatta-Brain_DM1919
Number of MERFISH genes	1000
Number of blank genes	119
Number of MERFISH bits	30
Number of sequential genes	3
Additional stain names	DAPI, PolyT, SLC17A7, MEF2C, APOE

vizgen

MERSCOPE Analysis Report (202412071006_Boulting-Slide5-NHPBrain-DM1919_VMSC21910)

Tuesday, December 10 at 10:31PM

Total transcripts detected: **142,420,325** Transcripts per 100 µm²: **81** Cell count: **378,166** Transcripts per cell – median: **198**

Overview Transcripts Cells Auxiliary stains

General

Statistics

Experiment area (mm²)	174.98
Total cell containing area (mm²)	169.51
Total cells detected	378,166
Total transcripts detected	142,420,325
Median transcripts per cell	198
Transcripts per 100µm² of transcript-containing tissue	82
Estimated tissue thickness (µm)	2.9
Tissue profile	exponential

Gene Panel Metadata

Gene panel serial number	3
Gene panel name	Mulatta-Brain_DM1919
Number of MERFISH genes	1000
Number of blank genes	119
Number of MERFISH bits	30
Number of sequential genes	3
Additional stain names	DAPI, PolyT, SLC17A7, MEF2C, APOE

Region folder structure

```
region_R1/
├── 202412041735_Boulting-Slide4-NHPBrain-DM1919_VMSC21910_region_R1.h5ad
├── 202412041735_Boulting-Slide4-NHPBrain-DM1919_VMSC21910_region_R1.vzg2
├── cell_boundaries.parquet
├── cell_by_gene.csv
├── cell_categories.csv
├── cell_metadata.csv
├── cell_numeric_categories.csv
├── detected_transcripts.csv
├── detected_transcripts.parquet
├── differentially_expressed_genes.csv
├── images/
└── summary.png
```

Pre-processed count matrix for entire region

Import into visualizer

Parquet stores data in columns, enabling efficient compression and faster queries

Pre-processed count matrix in csv format

Has entity ID (cell ID) in 1st column and leiden (cluster number) in 2nd column

Rows are cell ids (entity ID's) + columns of metadata stats for each cell (field of view or fov, x & y coordinates, transcript_count, etc)

Tif images of channels (DAPI, MEF2C, SLC17A7, ETC)

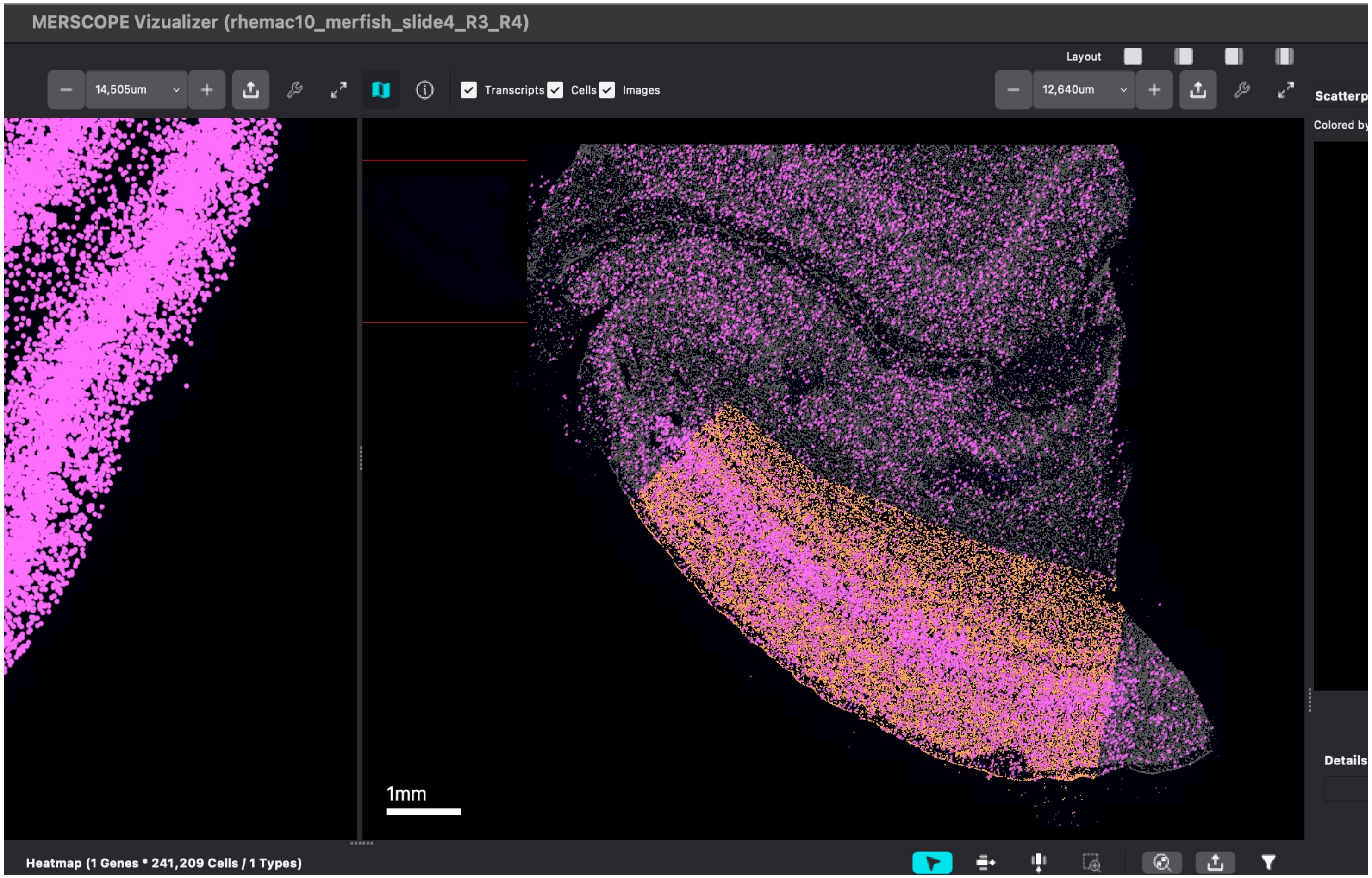
MERSCOPE Vizualizer

- Explore detected transcripts, and segmented cell boundaries
- Import single-cell clustering/annotation and dimensionality reduction results from external analysis
- Interact with UMAP/t-SNE scatter plots and heatmaps
- Select genes and create custom gene groups with the genes interface
- Select regions of interest for downstream analysis of your MERFISH data
- Goal: Visualize individual genes (EGR1, NPAS4, OSTN) to tell us if we see ocular dominance columns (ODCs) in tissue regions

[MERSCOPE® Vizualizer software]

Vizualizer example slide4 R3 OSTN

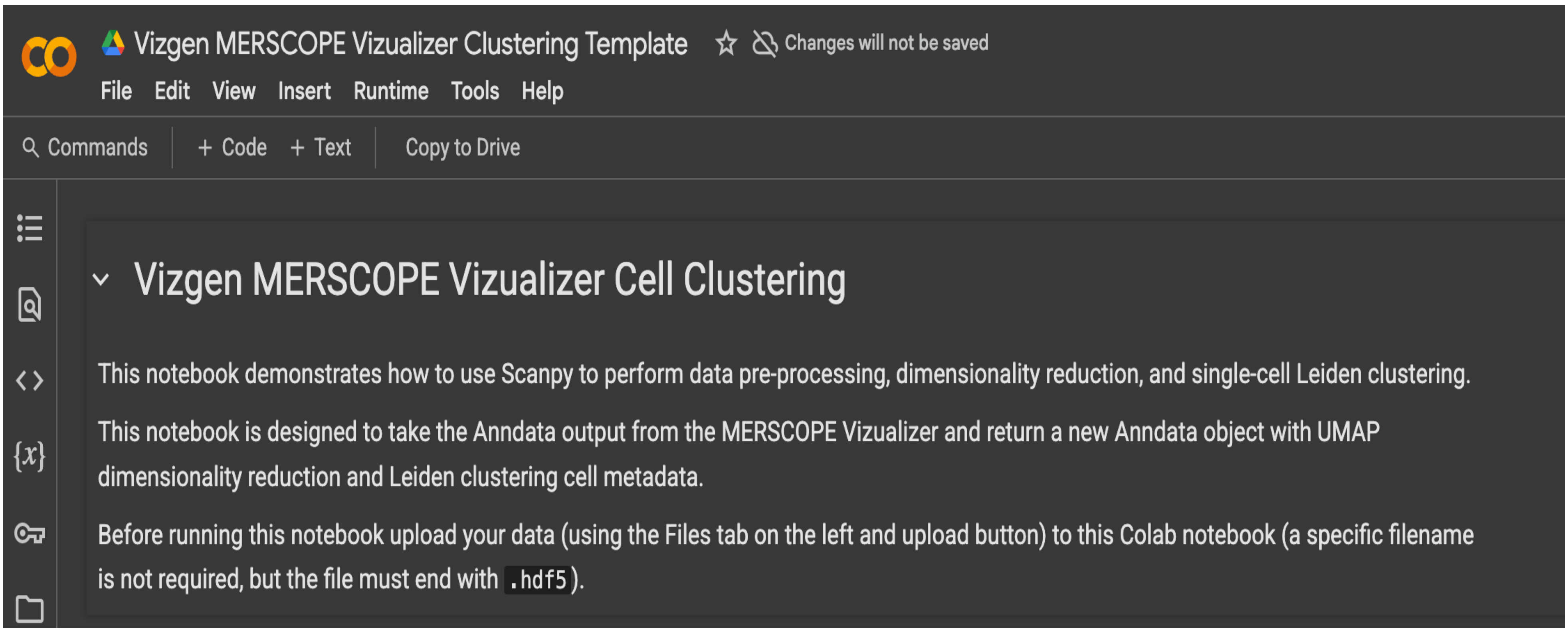
Pink = OSTN transcripts
Orange = Nuclei



Downstream analysis software

- Python
  - Vizgen officially supports
  - Scanpy
    - Used for scRNAseq/snRNAseq analysis (performs data pre-processing, dimensionality reduction, single-cell Leiden clustering)
    - Also performs analysis & visualization of spatial transcriptomics data
  - Squidpy
    - Used specifically for analysis & visualization of spatial transcriptomics data
- R
  - Currently, Vizgen doesn't officially support Seurat analysis yet
  - Seurat
    - Introduced extension October 2023 to analyze spatially-resolved RNA-seq data

Python analysis - Scanpy



Load Anndata object

- ad_viz = sc.read_h5ad("09-04-25_18-18_rhemac10_merfish_slide4_R3_Cells.hdf5")

```
View of AnnData object with n_obs × n_vars = 241209 × 881
    obs: 'Custom Cells groups', 'Datasets', 'volume', 'center_x', 'center_y'
    uns: 'Custom Cells groups_colors', 'Datasets_colors'
```

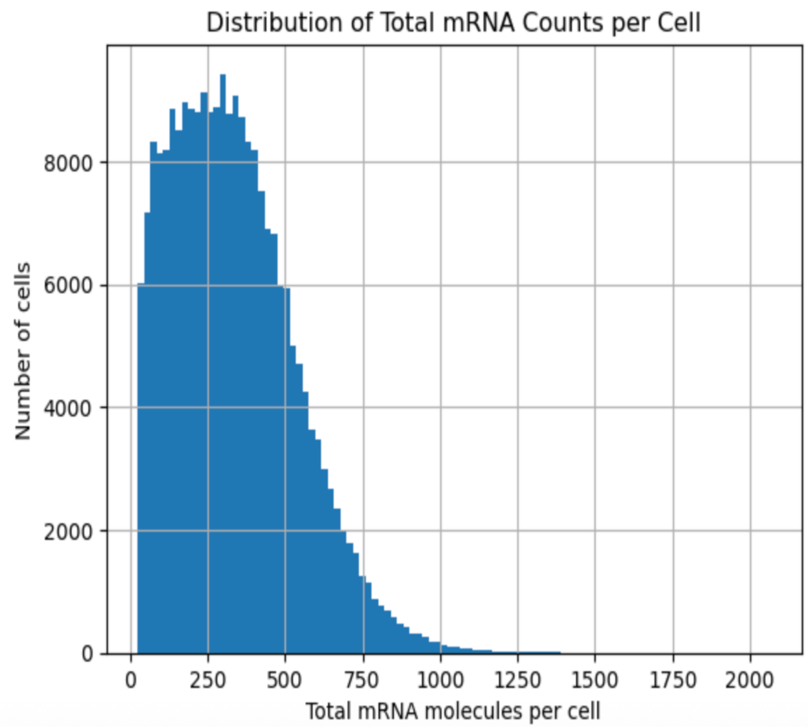
Filter Cells Based on Minimum Gene Expression Counts

- Filter the object by cells based on minimum gene expression
  - Keep cells with 25 mRNA transcripts or greater
- Cells with low total gene expression counts may indicate poor probe hybridization, low RNA capture efficiency, or other technical issues. Filtering out these cells is crucial to ensure that downstream analyses are based on reliable data

```
ad_viz = ad_viz[keep_cells]
# print object
ad_viz
# number of cells went from 241,209 to 236,965
```

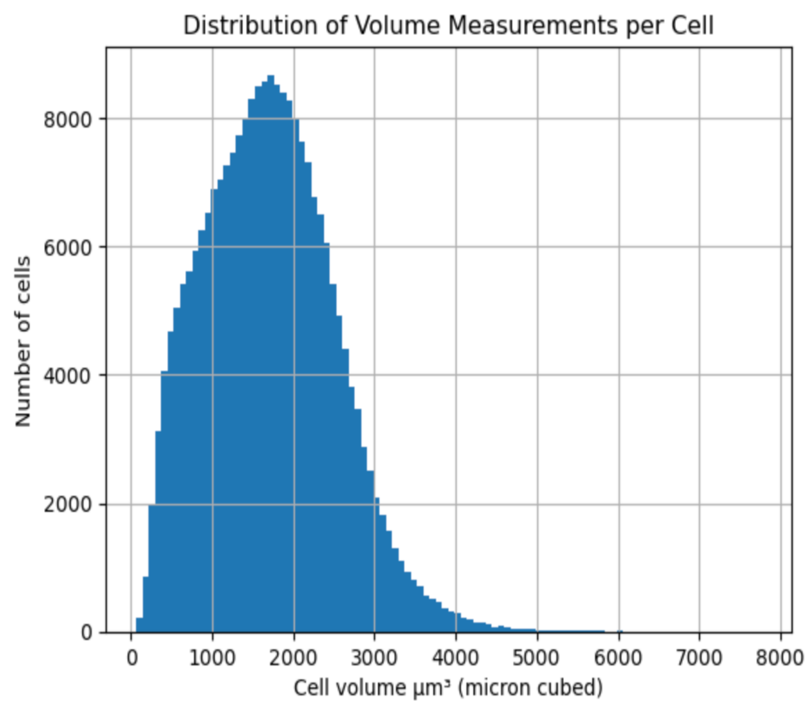
```
View of AnnData object with n_obs × n_vars = 236965 × 881
    obs: 'Custom Cells groups', 'Datasets', 'volume', 'center_x', 'center_y'
    uns: 'Custom Cells groups_colors', 'Datasets_colors'
```

```
241209-236965
# 4,244 cells removed
```



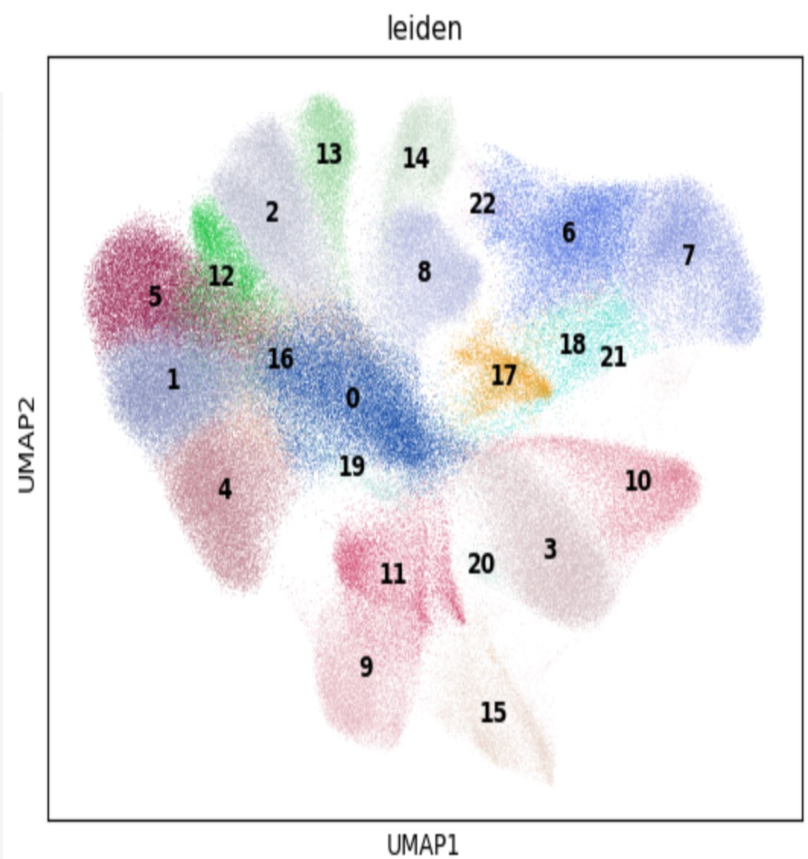
Filter Cells Based on Volume

- Cell volume is estimated physical size of each segmented cell in 3D space, usually measured in cubic microns (µm³)
- Filter cells based on volume
  - Keep cells that have volume of 50 or greater
- Comes from the cell segmentation step, where each cell is outlined in space based on detected transcripts and imaging data (vizualizer)
- Used to filter out poorly segmented cells, debris, or cells that are too small to be biologically meaningful
- Didn't remove any cells from ad_viz object



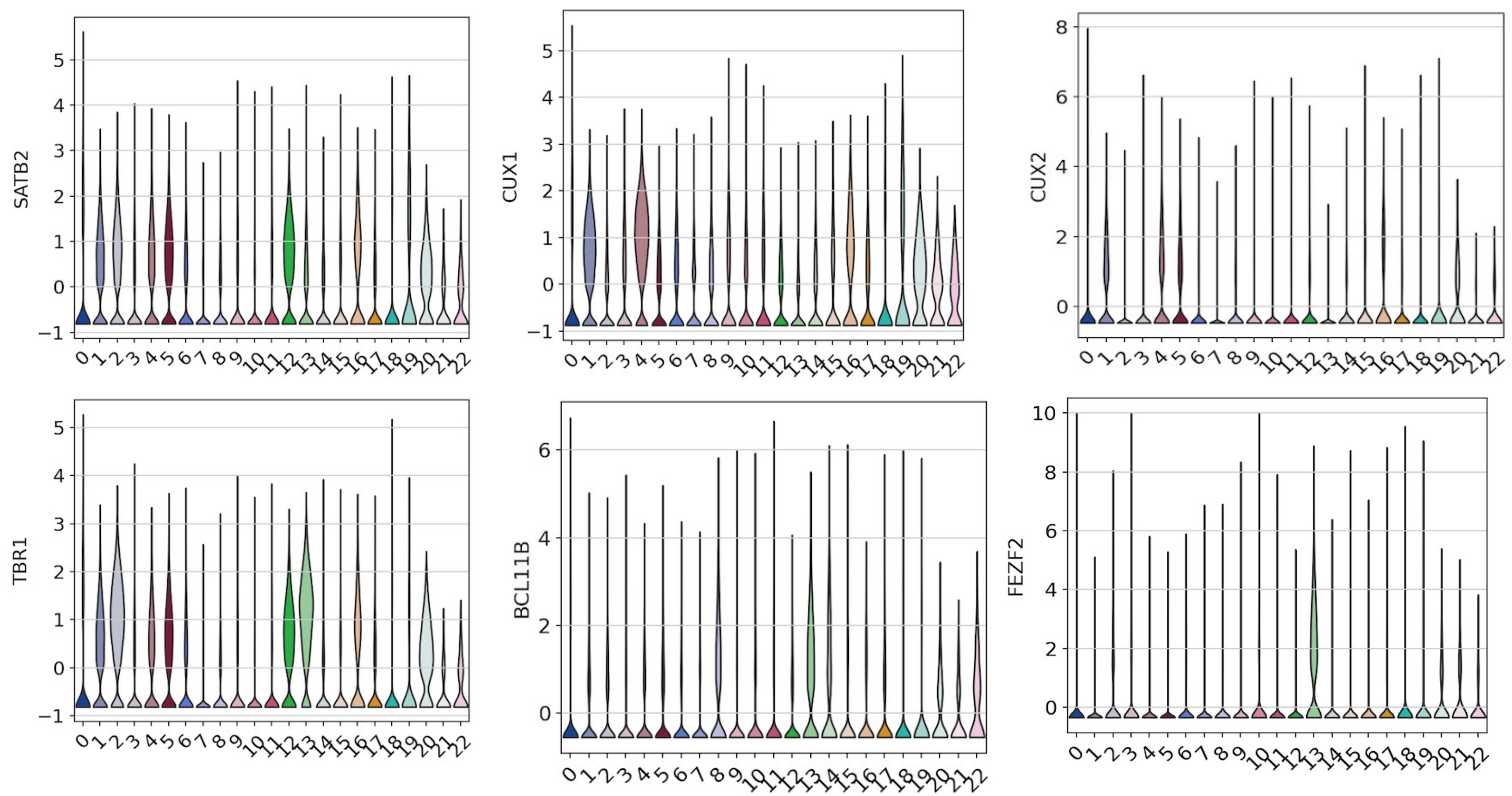
UMAP and Single-Cell Clustering

```
# Normalize each cell by total counts over all genes, so that every cell has the same total count after normalization
# pp means pre-processing
sc.pp.normalize_total(ad_viz)

# apply log transformation to normalized data
# log(X + 1) where X is the data matrix of shape n_obs(cells) x n_vars (genes)
sc.pp.log1p(ad_viz)

# Scale data to unit variance and zero mean
# max_value=10 caps extreme values to avoid distortion from outliers
sc.pp.scale(ad_viz, max_value=10)

# perform PCA on sparse matrix
# tl means tool
sc.tl.pca(ad_viz, svd_solver='arpack')

# find neighboors of cells (like the FindNeighbors function in Seurat!)
sc.pp.neighbors(ad_viz, n_neighbors=10, n_pcs=20)

# computes UMAP embedding for visualization
# based on neighborhood graph calculated above
sc.tl.umap(ad_viz)

# performs leiden clustering based on the resolution defined with = above
# Cluster cells using the Leiden algorithm [Traag et al., 2019], an improved version of the Louvain algorithm [Blonde
sc.tl.leiden(ad_viz, resolution=resolution)
```

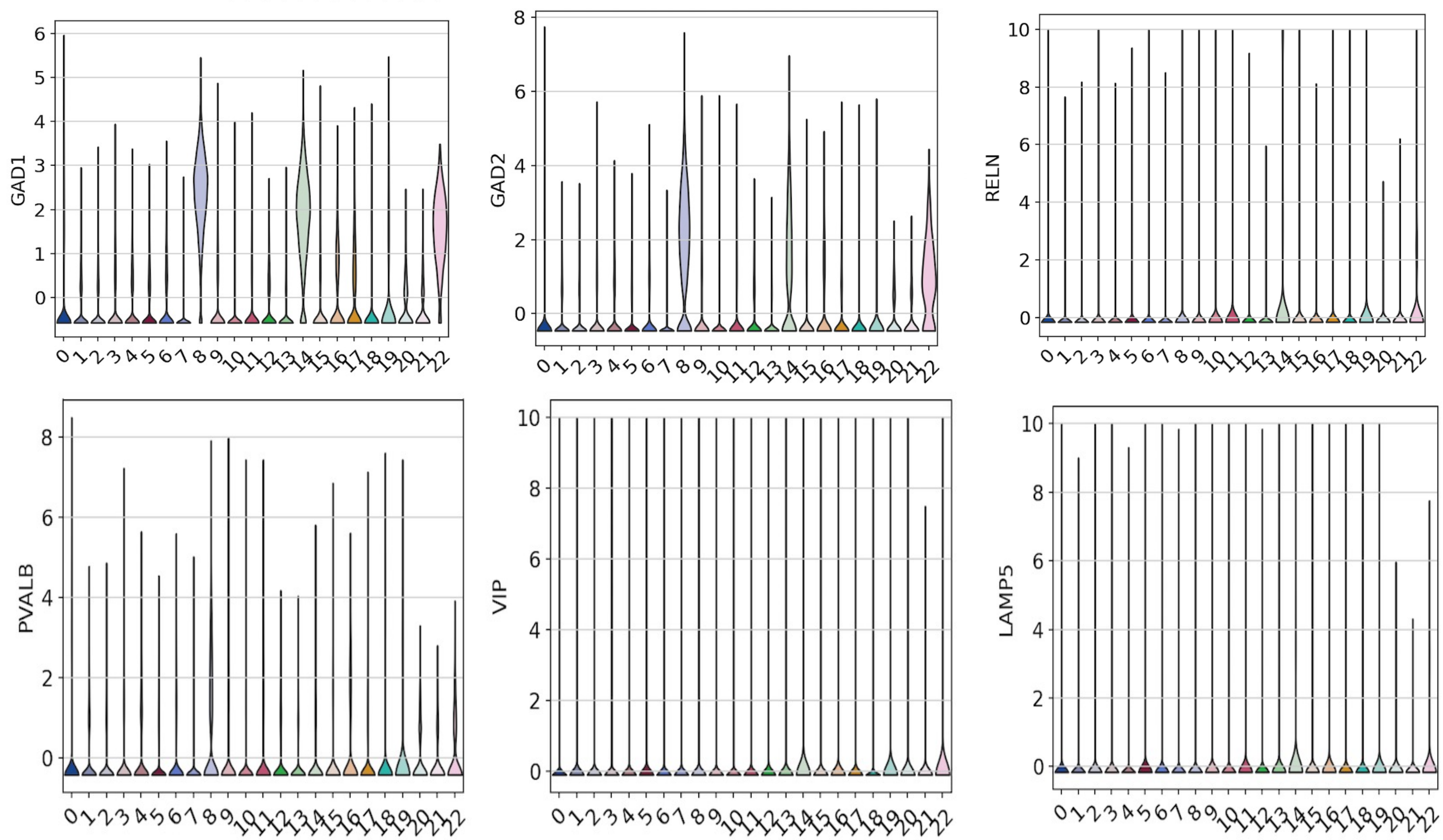


- PCs = 1-20
- Resolution = 1.5

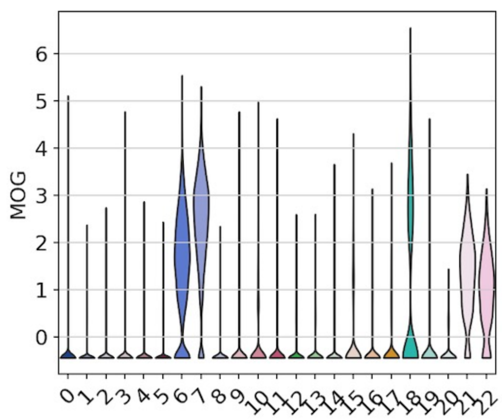
Gene violin plots – Excitatory clusters?

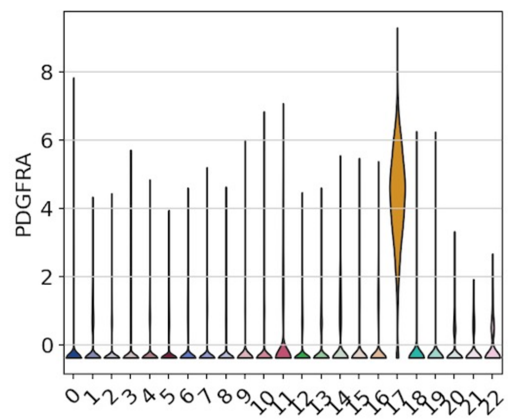


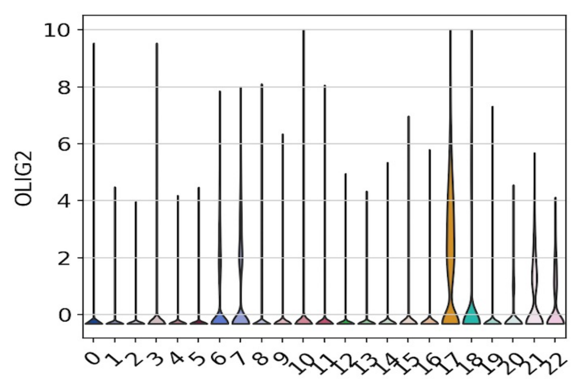
Gene violin plots – Inhibitory clusters?



Gene violin plots – other ?

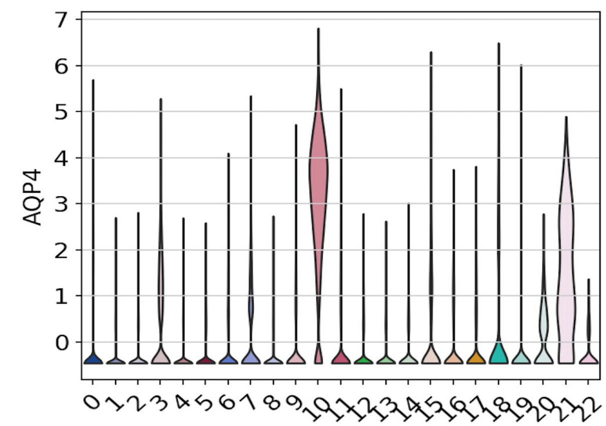




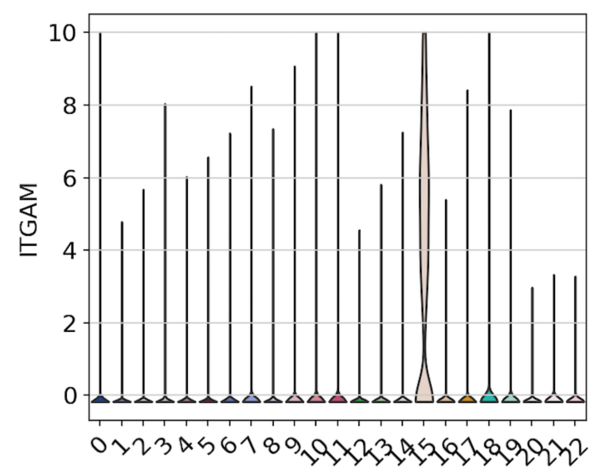


Oligodendrocytes

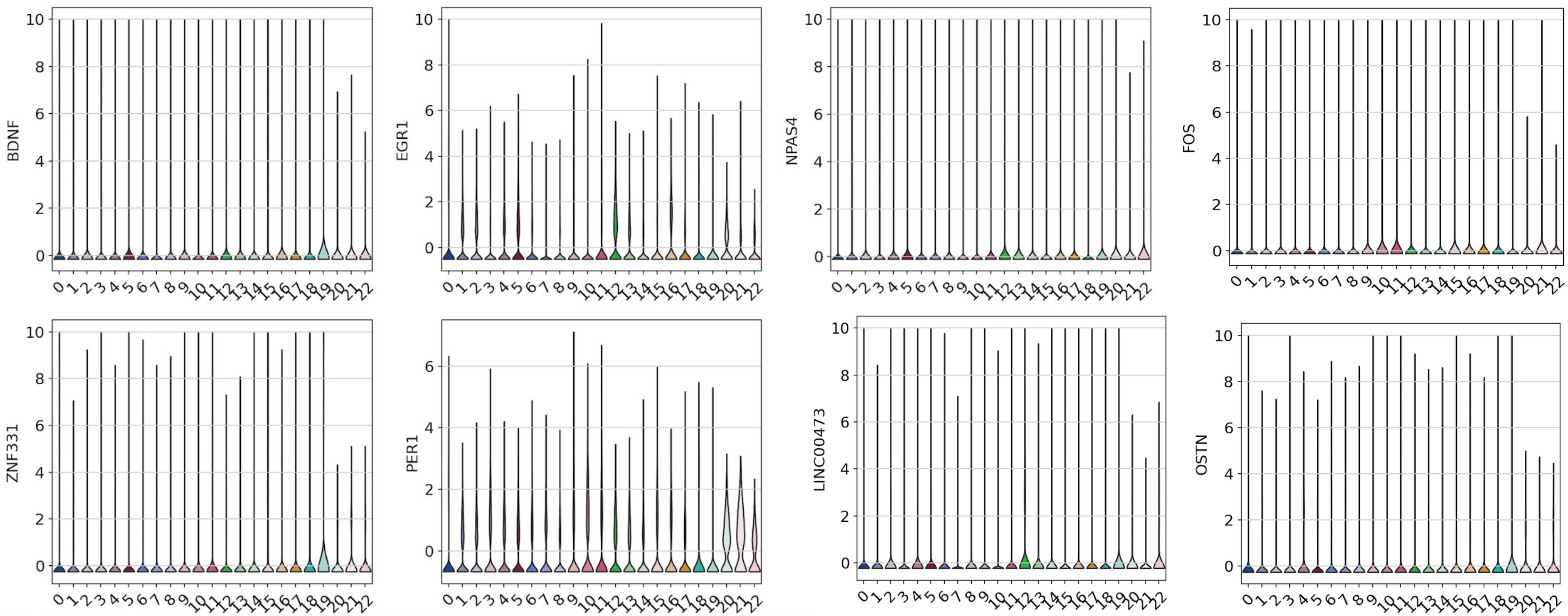
Astrocytes



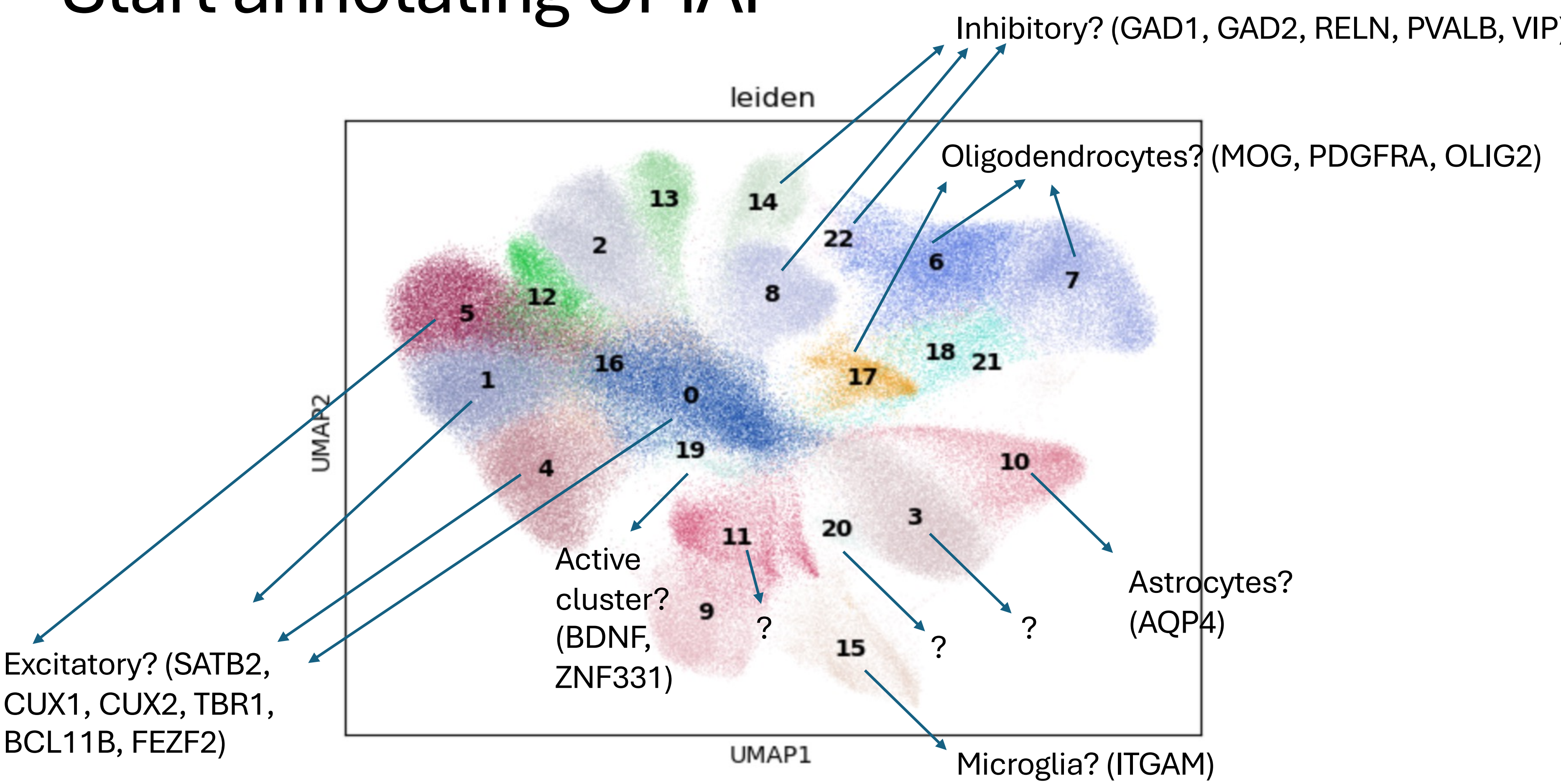
Microglia



Gene violin plots – Activity Dependent Genes



Start annotating UMAP



Progress

- Learning how to read in vzg2 files into MERSCOPE Vizualizer and view tissue regions to see if ODCs were captured
- Learning how to perform MERFISH analysis in python from exported hdf5 files from Vizualizer

Next steps

- View rest of tissue regions in slide 4 & 5 in Visualizer
- Export ROI's from Vizualizer (.hdf5) into python for downstream analysis